

CKD-Classification Assignment

1.Problem Statement:

- Stage1: Machine learning
- Stage2: Supervised learning
- Stage3: Regression

2.Rows & Columns:

- Total Number of columns: 28 columns
- Total Number of rows: 399 rows

3.Pre_processing Method:

- `dataset=pd.get_dummies(dataset,dtype=int,drop_first=True)`
- Given input from the database two datas are string type by using “get_dummies” function convert it into the numerical type.
- To avoid repeated information “drop_first=True” function to drop first columns.
- StandardScaler is the preprocessing function to convert the value from dataset to define range of value.

4.Final Good model with evaluation metric:

- LogisticRegression:

. The f1_macro value for best parameter{'penalty': 'l2', 'solver': 'lbfgs'}: 0.99168449

The confusion Matrix is:

```
[[45  0]
 [ 1 74]]
```

The Classification report is:

	precision	recall	f1-score	support
0	0.98	1.00	0.99	45
1	1.00	0.99	0.99	75
accuracy			0.99	120
macro avg	0.99	0.99	0.99	120
weighted avg	0.99	0.99	0.99	120

- Roc is: `np.float64(1.0)`

5.All Models:

- SupportVectorClassification:

The f1_macro value for best parameter{'gamma': 'auto', 'kernel': 'linear'}: 0.975:

The confusion Matrix is:

```
[[45  0]
 [ 3 72]]
```

The Classification report is:

	precision	recall	f1-score	support
0	0.94	1.00	0.97	45
1	1.00	0.96	0.98	75
accuracy			0.97	120
macro avg	0.97	0.98	0.97	120
weighted avg	0.98	0.97	0.98	120

np.float64(0.9994074074074074)

- DecisionTreeClassifier:

The f1_macro value for best parameter{}: {'criterion': 'gini', 'splitter': 'random'} 0.9751481

The confusion Matrix is:

```
[[45  0]
 [ 3 72]]
```

The Classification report is:

	precision	recall	f1-score	support
0	0.94	1.00	0.97	45
1	1.00	0.96	0.98	75
accuracy			0.97	120
macro avg	0.97	0.98	0.97	120
weighted avg	0.98	0.97	0.98	120

Roc_score is: 0.98

- RandomForest classifier

The f1_macro value for best parameter{}: {'criterion': 'gini', 'max_features': 'sqrt', 'n_estimators': 100} 0.9833:

The confusion Matrix is:

```
[[44  1]
 [ 1 74]]
```

The Classification report is:

	precision	recall	f1-score	support
0	0.98	0.98	0.98	45
1	0.99	0.99	0.99	75
accuracy			0.98	120
macro avg	0.98	0.98	0.98	120
weighted avg	0.98	0.98	0.98	120

np.float64(0.9997037037037035)

6. Here is the final result.

- While comparing all the models the evaluation metrics best in **Logistic Regression. The roc_score is np.float64(1.0)**